Impact of Enhanced SMC on Haemoglobin Trajectories in Children

Linear Mixed Models Analysis of a 6-Month Longitudinal RCT

Samuel Agyekum Philomene Cyubahiro Djadida Uwituze
Rabecca Kating'u Foumabn Marah Khadij Nouria

AIMS Rwanda | March 2026

Background & Research Question

The Problem

Malaria destroys red blood cells, causing severe anaemia in children across sub-Saharan Africa. Even after parasite clearance, iron deficiency persists.

Seasonal Malaria Chemoprevention(SMC) cuts malaria episodes by up to 75% during the rainy season, but does not address underlying iron deficiency.

Adding high-dose iron and nutritional supplements to SMC may protect haemoglobin levels safe with proper malaria prevention (Cochrane 2016).

Research Question

Do children receiving **Enhanced SMC** show significantly better haemoglobin trajectories over six months than children on **Standard SMC**?

Study Design & Dataset

Study Design

- 6-month longitudinal Randomised Controlled Trial
- **300 children** measured at 4 time points:
  Baseline, Month 2, Month 4, Month 6
- 1:1 randomisation:
  Standard SMC ($n = 155$) vs
  Enhanced SMC ($n = 145$)
- 1,200 total observations with no missing values

Key Variables

Role	Variable
Outcome	Haemoglobin concentration (g/dL)
Exposure	Treatment group (0=Standard, 1=Enhanced)
Time	Month (0, 2, 4, 6)
Covariates	Age at baseline (months); ITN use (0/1)
Key term	Treatment $\times$ Month interaction

Exploratory Analysis — Baseline Balance

Characteristic	**Standard** ($n = 155$)	**Enhanced** ($n = 145$)	p**-value**
Age at baseline (months)	34.4 ± 12.4	36.9 ± 13.3	0.094
Haemoglobin at baseline (g/dL)	11.98 ± 0.94	12.03 ± 0.96	0.668
ITN use — consistent (%)	63.9%	49.7%	**0.018**

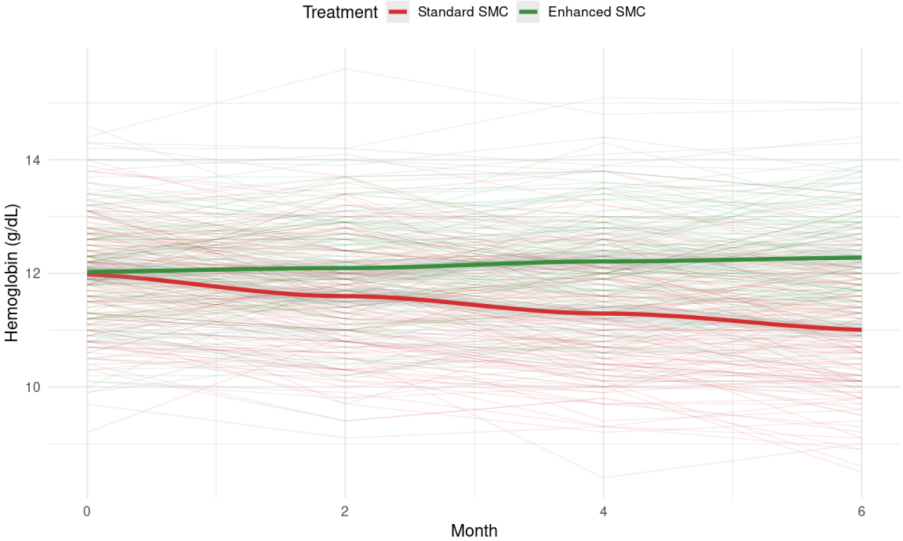
Randomisation Worked

Age and baseline haemoglobin were well-matched with neither group entered with a haemoglobin advantage.

ITN Imbalance Detected

Consistent ITN use was higher in Standard group (63.9% vs 49.7%, $p = 0.018$) which was included as covariate in all models.

Exploratory Analysis — Individual Haemoglobin Trajectories



Exploratory Analysis — Mean Haemoglobin Trajectories

Group	M0	M2	M4	M6
Standard SMC	11.98	11.60	11.29	11.00
Enhanced SMC	12.03	12.09	12.21	12.28

Mean haemoglobin (g/dL) by time point.
WHO threshold = 11 g/dL.

Key Observations

- **Standard SMC**: steady decline, reaches WHO threshold by Month 6
- **Enhanced SMC**: remains stable, slight increase throughout
- Groups start together and diverge from Month 2

Model Justification — Why Linear Mixed Models?

ICC

0.704

70.4% of haemoglobin variation is **between children**

Three Reasons to Use LMM

1. **Correlated data**: each child measured 4 times; standard regression assumes independence $\Rightarrow$ SEs too small
2. **High ICC (70.4%)**: most variation sits between children; ignoring this gives biased estimates
3. **Individual variability**: children differ in iron absorption, mosquito exposure, and treatment response

Model Building & Selection

Fixed Effects — AIC Comparison

Model	AIC
M1: Month only	2772.7
M2: + Treatment	2733.5
M3: + Age + ITN	2690.8
M4: + Interaction	**2470.5**

ΔAIC $= -220.3$ when interaction added

Random Effects Structure

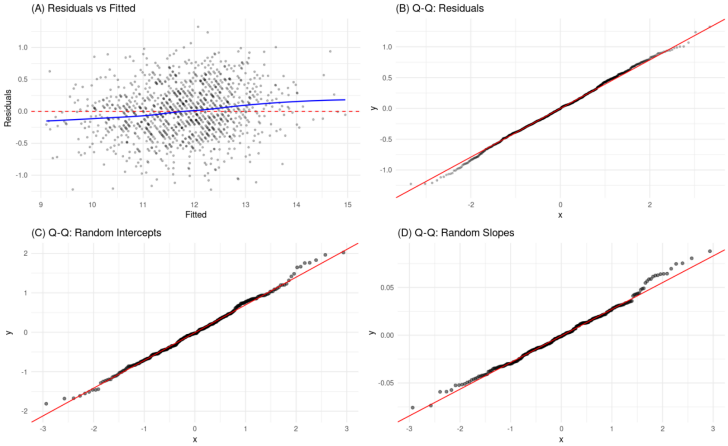
Model	AIC
Random Intercept only	2468.5
Intercept + Slope	**2463.2**

Likelihood Ratio Test: $\chi^2(2) = 9.27$, $p = 0.010$

Allowing each child their own rate of change fits significantly better biologically sensible (different iron absorption, exposure, response).

Final model: REML with Random Intercept + Random Slope for `month`, including `treatment` $\times$ `month` interaction.

Model Diagnostics



All four diagnostic panels pass so that means model assumptions are adequately met and inference is valid.

Results — Fixed Effects (REML)

Term	Estimate	SE	95% CI	p-value
Intercept	10.965	0.159	[10.653, 11.277]	< 0.001
Month	-0.162	0.010	$[-0.181, -0.143]$	< 0.001
Treatment (Enhanced vs Std)	0.078	0.101	$[-0.121,\ 0.276]$	0.443
Age at Baseline	0.021	0.004	$[\ 0.014,\ 0.028]$	< 0.001
ITN Use (Consistent)	0.430	0.097	$[\ 0.239,\ 0.620]$	< 0.001
Month × Treatment	**0.205**	**0.014**	**[0.177, 0.232]**	$< \mathbf{0.001}$

Key Finding: Month × Treatment Interaction

For every additional month, children on Enhanced SMC gained an extra **0.205 g/dL** compared to Standard SMC. Over 6 months this compounds to ≈**1.23 g/dL** which is clinically meaningful: large enough to move children from the anaemic into the healthy range (WHO threshold = 11 g/dL).

Discussion

Main Finding

The treatment-by-time interaction confirms groups diverged *progressively* and not a one-off difference. Iron supplementation accumulates its benefit gradually over months.

Clinical Significance

A 1.23 g/dL gain can shift children from anaemic to healthy status. Consistent ITN use independently added 0.43 g/dL, combining bed-nets with chemoprevention amplifies protection.

Limitations

- Simulated dataset, real data introduces missing visits, protocol deviations, unmeasured confounders (diet)
- Linear time assumption may miss acceleration or plateau in the treatment effect
- ITN use is binary self-report, known to overestimate actual usage
- 6-month window does not reveal whether haemoglobin advantage persists after the rainy season

Recommendation & Conclusion

Ministry of Health Recommendation

Give serious consideration to scaling up the Enhanced SMC programme nationally. Any rollout should include rigorous monitoring of implementation fidelity, cost-effectiveness, and longer-term haemoglobin outcomes beyond the rainy season.

0.205
g/dL per month
Extra haemoglobin gain per month (Enhanced vs Std)

$\sim$1.23
g/dL
Total advantage at Month 6

70.4%
ICC
Variation between children justifying LMM

Thank you